

jectory of a representative LFP cell and serve as the numerical reference for all subsequent experiments. The same trained parameter sets are then pruned, quantized, and exported to STM32 firmware. Identical feature streams ensure that accuracy, flash, RAM, latency, and the timing-derived energy proxy refer to the same stateful sequence. Figure 7.3 summarizes this end-to-end path from training to embedded benchmarking.

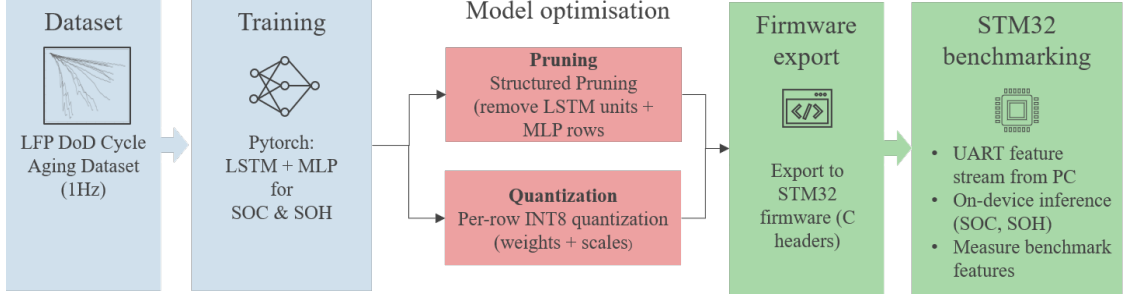


Figure 7.3: End-to-end deployment pipeline from the LFP ageing dataset and PyTorch training to pruning, quantization, C-header export, and final STM32 benchmarking. The same trained SOC and SOH parameter sets form the basis of the floating-point, pruned, and quantized deployments, which keeps the later comparisons numerically consistent.

7.4 Structured Pruning Configuration

The grouped magnitude criterion in Equation 3.21 and the local ranking principle in Equation 3.24 are instantiated here as one-shot structured post-training pruning. The implementation removes complete LSTM channels so that the exported tensors remain dense and the recurrent state becomes smaller. This structure directly matches the manually implemented C kernel and avoids an irregular element-wise sparsity mask. For hidden unit h , the implemented score combines the L2 norms of its input and recurrent rows across all four gates,

$$s_h = \sum_{g \in \{i, f, o, g\}} \left(\|W_{ih}^{(g)}[h, :]\|_2 + \|W_{hh}^{(g)}[h, :]\|_2 \right), \quad (7.11)$$

which follows the gate-grouping principle used in structured recurrent-network pruning.[47, 131, 132]

After sorting the scores, the retained index set $\mathcal{R}$ contains the H' highest-ranked units. The score itself uses only the input and recurrent gate rows. Once a channel is removed, the same index must also be removed from both LSTM bias vectors,